

CLOUD COMPUTING-II

UNIT-IV

Amazon Cloud Services

Amazon Web Services (AWS) is a leading cloud computing platform and a subsidiary of Amazon.com. It offers a vast and comprehensive set of cloud services, including computing power, storage, databases, machine learning, analytics, content delivery, and more. AWS is designed to provide scalable, flexible, and cost-effective solutions for businesses, startups, and individuals, allowing them to build, deploy, and manage applications and infrastructure without the need to invest in physical hardware or data centers.

Here are some key components and concepts to understand when introducing AWS:

1. **Global Infrastructure:** AWS operates data centers and Availability Zones in multiple regions worldwide. This global presence allows customers to host their applications and data closer to their end-users, improving latency and redundancy.
2. **Core Services:** AWS offers a wide range of core services, including:
 - ↗ **Compute:** Services like Amazon EC2 (Elastic Compute Cloud) for virtual servers and AWS Lambda for serverless computing.
 - ↗ **Storage:** Amazon S3 (Simple Storage Service) for object storage and Amazon EBS (Elastic Block Store) for block storage.
 - ↗ **Databases:** Amazon RDS (Relational Database Service), Amazon DynamoDB for NoSQL databases, and others.
 - ↗ **Networking:** Amazon VPC (Virtual Private Cloud) for network isolation, and services like Amazon Route 53 for DNS.
 - ↗ **Content Delivery:** Amazon CloudFront for content delivery.
 - ↗ **Analytics:** Amazon Redshift for data warehousing, Amazon EMR for big data processing, and Amazon Athena for querying data.
 - ↗ **Machine Learning:** AWS offers a suite of machine learning services, including SageMaker for model training and hosting, and Rekognition for image and video analysis.
3. **Security:** AWS provides robust security features, including IAM (Identity and Access Management) for access control, VPC for network security, and AWS Key Management Service (KMS) for encryption.
4. **Scalability:** AWS services are designed to scale up or down based on demand, ensuring that applications can handle traffic spikes and optimize costs during periods of low usage.
5. **Elasticity:** AWS allows you to automatically adjust resources to match your workload requirements, helping you avoid over-provisioning and underutilization of resources.
6. **Pay-as-You-Go Pricing:** AWS follows a pay-as-you-go pricing model, where you pay only for the resources and services you use. This flexibility can significantly reduce capital expenditures.
7. **Management Tools:** AWS offers various tools and services for managing your infrastructure, including AWS Management Console, AWS CLI (Command Line Interface), and AWS CloudFormation for infrastructure as code.

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8. **AWS Marketplace:** This is a platform where you can find and purchase pre-configured software and services to complement your AWS environment.
9. **Certifications:** AWS offers certification programs for individuals to validate their cloud expertise, including certifications for solutions architects, developers, and sysops administrators.
10. **Support and Documentation:** AWS provides extensive documentation, tutorials, and customer support options to help users get started and troubleshoot issues.

AWS has become a dominant player in the cloud computing industry, powering a wide range of applications and services across various sectors, including e-commerce, healthcare, finance, and more. Its flexibility, scalability, and rich ecosystem of services make it a popular choice for organizations looking to leverage the benefits of cloud computing.

AWS Web Components

It appears that you're interested in AWS web components. AWS provides a variety of web-related services and components that can be used to build and host web applications. Here are some of the key AWS web components:

1. **Amazon S3 (Simple Storage Service):** While primarily an object storage service, Amazon S3 is commonly used to host static websites. You can configure an S3 bucket to serve as a web host, enabling you to store HTML, CSS, JavaScript, images, and other assets for your website.
2. **Amazon EC2 (Elastic Compute Cloud):** EC2 instances can be used to host web applications and websites. You can launch virtual servers with varying compute capacity to run web servers like Apache or Nginx, along with your application code.
3. **Amazon CloudFront:** CloudFront is a Content Delivery Network (CDN) service that distributes your web content to edge locations worldwide. It accelerates content delivery by caching static and dynamic content, reducing latency for end-users.
4. **AWS Elastic Beanstalk:** Elastic Beanstalk is a Platform as a Service (PaaS) offering that simplifies the deployment and management of web applications. It supports multiple programming languages and web frameworks, allowing you to focus on your code while AWS handles infrastructure management.
5. **AWS Amplify:** Amplify is a set of tools and services that helps frontend developers build full-stack web and mobile applications. It provides features like authentication, APIs, and hosting, making it easier to develop and deploy web applications.
6. **Amazon API Gateway:** If you're building RESTful APIs for your web applications, Amazon API Gateway can help you create, publish, and manage APIs at scale. It can be used in conjunction with AWS Lambda for serverless backend services.
7. **AWS Lambda:** Lambda allows you to run code in response to events without provisioning or managing servers. You can use Lambda to create serverless backend functions for your web applications.

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8. **Amazon RDS (Relational Database Service):** RDS provides managed database services for web applications. It supports multiple database engines like MySQL, PostgreSQL, and SQL Server, making it easy to set up and operate databases.
9. **Amazon DynamoDB:** If you require a NoSQL database for your web application, DynamoDB is a fully managed, highly scalable option. It's designed for low-latency, high-throughput applications.
10. **Amazon Route 53:** Route 53 is AWS's scalable Domain Name System (DNS) web service. It allows you to register domain names and route traffic to various AWS resources, including S3 buckets, EC2 instances, and load balancers.
11. **Amazon Cognito:** Cognito provides user identity and access management for web and mobile applications. It helps you add authentication, authorization, and user management features to your web apps.
12. **AWS App Runner:** This service simplifies the deployment of containerized web applications. You provide your source code, and AWS App Runner automatically builds and deploys your application.

These are just a few of the many AWS components and services that can be used to build and host web applications. AWS provides a flexible and scalable platform for web development, allowing you to choose the right services and resources to meet your application's requirements.

The AWS console

The AWS Management Console is a web-based interface provided by Amazon Web Services (AWS) that allows users to interact with and manage their AWS resources and services. It serves as a central hub for controlling and configuring various AWS offerings. Here are some key points about the AWS Management Console:

1. **Access:** To access the AWS Management Console, users need AWS credentials, typically an Access Key ID and Secret Access Key, which can be obtained by creating an AWS account. Users can also log in using their AWS Identity and Access Management (IAM) credentials if they have been granted access.
2. **Dashboard:** The console's landing page features a customizable dashboard where users can add widgets and view important information about their AWS resources, such as billing and usage, recent activity, and service health notifications.
3. **Services:** AWS provides a wide range of services, and the console organizes them into categories, making it easier for users to find and manage specific services. Some common categories include Compute, Storage, Databases, Networking, Security & Identity, and Analytics.
4. **Navigation:** The console offers a left-hand navigation pane that lists AWS services by category. Users can click on a service to access its specific features and settings.

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5. **Search:** There is a search bar at the top of the console that allows users to quickly find AWS resources and services by name or keyword. This is especially useful when you have many resources to manage.
6. **Resource Management:** Users can create, configure, and delete AWS resources through the console. This includes launching virtual machines (EC2 instances), creating databases (RDS instances), setting up storage buckets (S3), and more.
7. **Monitoring and Logging:** The console provides access to AWS CloudWatch for monitoring AWS resources and services, as well as AWS CloudTrail for tracking and logging AWS API activity.
8. **Identity and Access Management (IAM):** Administrators can use the console to manage IAM users, groups, roles, and permissions, controlling who can access specific AWS services and resources.
9. **Billing and Cost Management:** The console includes tools for monitoring your AWS costs and usage, setting up billing alerts, and accessing detailed billing reports.
10. **Support and Documentation:** Users can access AWS support options, documentation, and FAQs directly from the console to get help with any issues or questions they may have.
11. **Customization:** Users can customize the console by rearranging and pinning frequently used services to the dashboard for easy access. Additionally, they can apply themes for a personalized look and feel.
12. **Mobile App:** AWS also provides a mobile app that offers limited console functionality, allowing users to monitor resources and perform some basic management tasks from their mobile devices.

The AWS Management Console is a user-friendly interface designed to make it easier for users to interact with and manage AWS resources and services. However, for more complex and automated tasks, AWS also offers command-line tools and APIs for programmatic access and automation of AWS resources and services.

AWS compute

In cloud computing, the term "compute" refers to the processing power, memory, storage, networking, and other resources required for the computational success of any program aws.amazon.com. AWS offers a range of compute services that provide secure and resizable compute capacity in the cloud. These services allow you to launch applications when required without upfront commitments aws.amazon.com.

AWS Compute services include various instance types optimized for different use cases. They comprise varying combinations of CPU, memory, storage, and networking capacity and give you the flexibility to choose the appropriate mix of resources for your applications docs.aws.amazon.com.

Here are a few types of instances:

- **Compute optimized instances:** These instances are used to run high-performance compute applications that require fast network performance, extensive availability, and high input/output (I/O) operations per second aws.amazon.com.

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- ↯ Accelerated computing instances: They use hardware accelerators, or co-processors, to perform functions, such as floating point number calculations, graphics processing, or data pattern matching, more efficiently than is possible in software running on CPUs
aws.amazon.com.
- ↯ Memory optimized instances: These instances use high-speed, solid-state drive infrastructure to provide ultra-fast access to data and deliver high performance. They are ideal for applications that require more memory and less CPU power aws.amazon.com.
- ↯ Storage Optimized instances: These instances are optimized for applications that require high sequential I/O performance, disk throughput, and low cost storage for very large data sets
aws.amazon.com.

AWS Compute also offers various pricing models, including On-Demand pricing, Spot Instances, and Savings Plans.

- ↯ On-Demand Pricing: You pay only for the compute you need with no long-term commitments aws.amazon.com.
- ↯ Spot Instances: Spot Instances allow you to take advantage of unused EC2 capacity in the AWS cloud and are available at a discount of up to 90% compared to On-Demand prices
aws.amazon.com.
- ↯ Savings Plans: With Savings Plans, you can save up to 72% on AWS compute usage aws.amazon.com.

AWS Compute services are designed to optimize costs, as you pay only for the instance or resource you need, for as long as you use it, without requiring long-term contracts or complex licensing aws.amazon.com. AWS also offers a Free Tier that allows users to explore products for free aws.amazon.com.

Amazon Elastic Compute Cloud (Amazon EC2) is a web service provided by Amazon Web Services (AWS) that offers scalable and resizable compute capacity in the cloud. EC2 enables users to run virtual machines, known as instances, on the AWS infrastructure. It is one of the core services within AWS and is widely used for various computing tasks, including web hosting, application deployment, data processing, machine learning, and more.

Here are some key features and aspects of Amazon EC2:

1. **Scalability:** EC2 allows you to easily scale your compute capacity up or down based on your application's needs. You can launch multiple instances as needed and terminate them when they are no longer required.
2. **Variety of Instance Types:** EC2 offers a wide range of instance types optimized for different use cases. These instance types vary in terms of CPU, memory, storage, and GPU capacity. Users can choose the most suitable instance type for their specific workloads.
3. **Operating System Support:** EC2 instances can run various operating systems, including Linux, Windows, macOS, and other specialized OS options. Users can select the desired OS and configure it as needed.

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4. **Elastic Load Balancing:** EC2 can be integrated with Elastic Load Balancing to distribute incoming traffic across multiple instances, improving application availability and fault tolerance.
5. **Security:** AWS provides various security features for EC2, such as Virtual Private Cloud (VPC) for network isolation, Security Groups for firewall rules, and Identity and Access Management (IAM) for fine-grained access control.
6. **Data Storage:** EC2 instances can be attached to different types of storage, including Amazon Elastic Block Store (EBS) for block storage and Amazon S3 for object storage. This allows you to manage your data separately from your instances.
7. **Auto Scaling:** AWS Auto Scaling can be used in conjunction with EC2 to automatically adjust the number of running instances based on predefined policies, ensuring optimal performance and cost efficiency.
8. **Spot Instances:** EC2 offers Spot Instances, which allow users to bid on spare AWS capacity at significantly reduced prices. This is a cost-effective option for workloads that can tolerate interruptions.
9. **Reserved Instances:** Users can purchase Reserved Instances to reserve capacity for a specific duration, which often results in cost savings compared to on-demand pricing.
10. **Instance Images:** EC2 instances can be saved as Amazon Machine Images (AMIs), allowing you to create and launch new instances with the same configuration and applications.
11. **Monitoring and Management:** AWS provides various tools like Amazon CloudWatch for monitoring and AWS Systems Manager for managing EC2 instances, making it easier to maintain and troubleshoot your infrastructure.

Overall, Amazon EC2 is a versatile and flexible cloud computing service that provides the building blocks for deploying and managing virtual servers in the AWS cloud, making it a fundamental component of many modern cloud-based applications and services.

EC2 instance using the AWS Management Console,

follow these steps:

1. Open the Amazon EC2 console.
2. Choose Launch Instance.
3. Choose an Amazon Machine Image (AMI). The AMI provides the operating system, application server, and applications for your instance.
4. Choose an instance type. The instance type determines the hardware configuration and size of your instance.

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5. Configure instance details. This includes specifying the number of instances, the key pair to use, and the network settings.
6. Add storage. You can add one or more volumes to your instance.
7. Add tags. Tags are metadata that you can use to organize and manage your resources.
8. Configure security group. A security group is a set of firewall rules that control inbound and outbound traffic to your instance.
9. Review instance launch and select key pair.
10. Choose Launch Instances.

It can take a few minutes for your instance to launch and become available. Once your instance is running, you can connect to it using a SSH client.

Here are some tips for launching an EC2 instance:

- ↗ Choose the right AMI. There are a wide variety of AMIs available, including AMIs for different operating systems, applications, and development platforms.
- ↗ Choose the right instance type. The instance type that you choose will depend on your workload and budget.
- ↗ Configure instance details. Make sure to specify the number of instances, the key pair to use, and the network settings.
- ↗ Add storage. You can add one or more volumes to your instance, depending on your storage needs.
- ↗ Add tags. Tags are a good way to organize and manage your resources.
- ↗ Configure security group. Make sure to configure the security group to allow inbound and outbound traffic to your instance.

If you have any questions or need help launching an EC2 instance, you can contact AWS Support.

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Amazon storage system

Amazon Web Services (AWS) offers a variety of storage services to meet different needs and use cases. These services are designed to provide scalable, durable, and highly available storage solutions for applications and data in the cloud. Here are some of the key storage services provided by AWS:

1. **Amazon S3 (Simple Storage Service)**:

- Amazon S3 is an object storage service that allows you to store and retrieve data, such as files, documents, images, and videos, in a highly durable and available manner.

- It offers flexible storage classes with different performance and cost characteristics, including Standard, Intelligent-Tiering, Glacier, and more.

- S3 provides features like versioning, data replication, and access control to help manage and secure your data.

2. **Amazon EBS (Elastic Block Store)**:

- Amazon EBS provides block storage volumes that can be attached to EC2 instances, offering persistent and high-performance storage.

- EBS volumes come in different types, including General Purpose (SSD), Provisioned IOPS (SSD), and Magnetic (HDD), each optimized for specific workloads.

- EBS snapshots enable you to back up your volumes and create point-in-time copies for disaster recovery.

3. **Amazon EFS (Elastic File System)**:

- Amazon EFS is a managed file storage service that can be mounted to multiple EC2 instances, enabling shared access to files and data across instances.

- It is designed for use cases like content management, web serving, and data sharing among applications.

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- EFS is highly available and scalable, automatically growing and shrinking as you add or remove files.

4. **Amazon RDS (Relational Database Service)**:

- Amazon RDS is a managed database service that supports various database engines, including MySQL, PostgreSQL, Oracle, and SQL Server.

- It provides scalable and highly available database storage with automated backups and maintenance.

- RDS takes care of routine database tasks like patching, backups, and scaling, allowing you to focus on your application logic.

5. **Amazon DynamoDB**:

- Amazon DynamoDB is a managed NoSQL database service designed for fast and scalable applications.

- It offers high-performance, low-latency data access with automatic scaling and built-in security features.

- DynamoDB is suitable for use cases like mobile and gaming applications, real-time analytics, and IoT data storage.

6. **Amazon Glacier**:

- Amazon Glacier is a low-cost archival storage service for data that is infrequently accessed.

- It is designed for long-term data retention and compliance requirements.

- Glacier provides multiple retrieval options with varying retrieval times to meet different needs.

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7. ****Amazon S3 Glacier (formerly Glacier Deep Archive)**:**

- S3 Glacier is an even lower-cost archival storage option that integrates with Amazon S3 for seamless data archiving.

8. ****Amazon Storage Gateway**:**

- AWS Storage Gateway is a hybrid cloud storage service that enables on-premises applications to seamlessly use AWS cloud storage.

- It supports file, volume, and tape gateways to connect your on-premises environments to AWS storage services.

These are some of the core storage services offered by AWS. Depending on your specific requirements, you can choose the most appropriate storage service or combination of services to meet your application's needs for scalability, durability, and cost-effectiveness in the cloud.

Amazon database services

Amazon Web Services (AWS) provides a wide range of database services to cater to various data storage, processing, and management needs. These managed database services are designed to simplify database administration, enhance scalability, and improve reliability. Here are some of the key Amazon database services:

1. ****Amazon RDS (Relational Database Service)**:**

- Amazon RDS is a managed relational database service that supports popular database engines such as MySQL, PostgreSQL, MariaDB, Oracle, and Microsoft SQL Server.

- It automates routine database tasks like patching, backups, and scaling, allowing you to focus on your application.

- Amazon RDS provides high availability, read replicas, and automated backups for data durability.

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2. **Amazon Aurora**:

- Amazon Aurora is a fully managed, high-performance relational database engine that is compatible with MySQL and PostgreSQL.
- It offers greater performance, scalability, and durability compared to standard MySQL or PostgreSQL databases.
- Aurora provides features like automatic failover, continuous backup, and replication across multiple Availability Zones.

3. **Amazon DynamoDB**:

- Amazon DynamoDB is a managed NoSQL database service designed for fast and scalable applications.
- It offers low-latency data access, automatic scaling, and built-in security features.
- DynamoDB is suitable for use cases like mobile and gaming applications, real-time analytics, and IoT data storage.

4. **Amazon DocumentDB**:

- Amazon DocumentDB is a managed NoSQL database service compatible with MongoDB.
- It provides the scalability and performance of a document database while offering the reliability and durability of Amazon Aurora.
- DocumentDB is ideal for applications that require a flexible schema and JSON-like data structures.

5. **Amazon Redshift**:

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-Amazon Redshift is a fully managed data warehousing service that is designed for analytical workloads.

-It enables you to query and analyze large datasets using standard SQL and integrates with popular business intelligence tools.

- Redshift offers fast query performance, columnar storage, and data compression.

6. **Amazon Neptune**:

-Amazon Neptune is a fully managed graph database service that supports two popular graph models: Property Graph and RDF (Resource Description Framework).

-It is suitable for building applications that require graph analytics, social networking, recommendation engines, and knowledge graphs.

7. **Amazon ElastiCache**:

-Amazon ElastiCache is a managed in-memory caching service that supports popular caching engines like Redis and Memcached.

-It helps improve application performance by reducing database load and providing low-latency access to frequently accessed data.

8. **Amazon QLDB (Quantum Ledger Database)**:

-Amazon QLDB is a fully managed ledger database service designed for applications that require an immutable and cryptographically verifiable transaction log.

-It is ideal for use cases like supply chain management, financial auditing, and compliance.

These Amazon database services cater to a wide range of use cases and data storage requirements, from traditional relational databases to NoSQL databases, data warehousing, graph databases, and ledger databases. Users can choose the

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appropriate database service based on their application's needs for scalability, data model, and performance characteristics.

Running programs written with the AWS flow framework for Java

Running programs written with the AWS Flow Framework for Java involves the coordination of workflows and activities using Amazon Simple Workflow (SWF). Here's a step-by-step guide on how to run programs written with the AWS Flow Framework for Java:

1. Setup Your Development Environment:

- Ensure you have Java installed on your development machine.
- Set up the AWS SDK for Java by including the necessary libraries and configuring your AWS credentials using AWS Access Key ID and Secret Access Key.

2. Write Your Workflow and Activity Code:

- Create Java classes for your workflow and activities. Workflows define the coordination logic, while activities represent individual tasks.
- Annotate your workflow and activity methods with the **@Asynchronous** annotation to indicate that they are asynchronous.

3. Define Workflow and Activity Workers:

- Create worker classes for your workflows and activities. These worker classes implement the corresponding interfaces generated by the AWS Flow Framework.
- Worker classes poll Amazon SWF for tasks and execute them when assigned.

4. Register Workflow and Activity Types:

- Use the **@WorkflowRegistrationOptions** and **@ActivityRegistrationOptions** annotations to specify the registration options for your workflow and activity types.

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- ✎ You can also use the **RegisterWorkflowTypes** and **RegisterActivityTypes** command-line tools provided by the AWS Flow Framework to register types.

5. Start a Workflow Execution:

- ✎ To start a workflow execution, create an instance of your workflow interface and use it to initiate the workflow. Provide any required input parameters.

```
MyWorkflowClientExternalFactory factory = new MyWorkflowClientExternalFactoryImpl();
```

```
MyWorkflowClientExternal client = factory.getClient();
```

```
Promise<Void> workflowExecution = client.myWorkflowMethod(input);
```

1. Workflow Execution and Coordination:

- ✎ The AWS Flow Framework takes care of orchestrating the execution of activities and managing the workflow's state. It schedules and executes activities based on your coordination logic.

2. Activity Execution:

- ✎ Activity workers poll Amazon SWF for tasks and execute the activities when assigned. Activities should be implemented in the corresponding activity worker classes.

3. Handle Failures and Retries:

- ✎ You can define retry policies for activities and implement error handling within your workflow and activity methods to ensure robustness.

4. Completion and Results:

- ✎ Activities and workflow steps return results that can be used in subsequent steps or to determine the overall outcome of the workflow.

5. Monitoring and Management:

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- ↗ You can use the Amazon SWF console and Amazon CloudWatch for monitoring the progress and health of your workflows. Additionally, the AWS SDK for Java provides logging and debugging capabilities.

6. Terminate or Cancel Workflows:

- ↗ You can manually terminate or cancel workflow executions using the AWS Management Console or programmatically using the AWS SDK for Java.

7. Scaling:

- ↗ You can scale the number of workflow and activity workers to handle increased workloads by configuring and launching multiple worker instances.